

Lab 6

(25+25 = 50 points total)

1. The inner product of two vectors, x and y , both of length n , is defined as:

$$\sum_{i=1}^n x_i y_i$$

For example, the inner product of $x = [2,4,6]$ and $y = [1,2,3]$ is $2 \times 1 + 4 \times 2 + 6 \times 3 = 28$. In this assignment, you will write an ARM assembly code to compute the inner product of two vectors. Put the following data snippet in your code:

```
.data:
array_1:      .word 1, 3, 5, 7, 9, 11
array_2:      .word 2, 4, 6, 8, 10, 12
innerProduct: .word 0
arrayLength:  .byte 6
```

Your code should load the two arrays into registers, compute the inner product, then store it into `innerProduct`. **You must use a loop. You cannot use any other variable in memory other than the ones shown in the snippet above.**

2. The Hamming distance between two bitstreams is defined as the number of positions in which the bitstreams differ. For example, the Hamming distance between 1010 (hex A) and 1100 (hex C) is 2 since the two bitstreams differ in the 1st and 2nd bit positions. The Hamming distance between 0xAA and 0xCC is therefore $2 \times 2 = 4$ and the Hamming distance between 0xAAAAAAAA and 0xCCCCCCCC is $8 \times 2 = 16$.

In this assignment, you will compute the Hamming distance between two numbers `x` and `y` and write the result in memory (variable `HammDist` in the code snippet below).

Put the following snippet in your program and proceed from there. **You must use a loop in your code. You cannot use any other variable in memory other than the ones shown in the snippet below. This problem might appear hard, but it requires just 20 lines of instruction, if you think carefully about how to solve it.**

Here are some Hamming distance values you can use to test your code:

`HD(011100110, 101011111) = 6`, `HD(01110011011101001, 10101111100100001) = 9`

`HD(0111001101110, 1010111110010) = 8`

```
_data:
x: .word 0xAA
y: .word 0xCC
HammDist: .word 0
```

Note: You can work on the lab assignment during the week and just demo it during the lab section next week (the week of June 1). Or you can choose to work on it during the lab hours next week. If you choose the latter option, please note that no grace period will be provided to finish your lab.